

ROYDAN CRUZ

Methuen, Ma | roydancruz7@gmail.com | <http://www.linkedin.com/in/roydan-cruz-a93a58193> | [Data Portfolio](#)

EDUCATION

Boston College,

M.S in Data Science,

Cumulative GPA: 3.84

Chestnut Hill, MA

August 2026

- Relevant courses: Natural Language Processing, Database Systems and Data Preparation, Applied AI and Machine learning, Deep Learning and AI, Data Visualization and Storytelling

UMASS Lowell, University of Massachusetts Lowell

B.A in Economics

Cumulative GPA: 3.92

Lowell, MA

December 2022

- Honors / Awards: Summa Cum Laude, Omicron Delta Epsilon (ODE) International Honor Society for Economics member, MIT IDDEAS Scholar

RELEVANT COURSE PROJECTS

Data Engineering – Collaborative Capstone Project

Database Design for Climate-Resilient Water Management Systems in Tanzania

Dec 2025

Streamlit Link: <https://climate-resilient-database.streamlit.app/>

- Created a spatial database in PostGIS to centralize disparate environmental, demographic, water usage, and water accessibility data.
- Established ETL pipelines in Python to convert raw rasterized environmental datasets into regional tabular statistics to load into PostGIS.
- Created a Streamlit dashboard to highlight low-water resilient areas in Tanzania supporting targeted water grid investment.
- Created materialized views reduce querying latency during downstream k-means clustering analysis and regular views to power the Streamlit dashboard.

Computer Vision

Brain Cancer Medical Imaging Classification

Mar 2025

GitHub Link: <https://github.com/ApollocreedXI/Brain-Cancer-Medical-Imaging-Classification/tree/main>

- Employed deep convolutional neural network architectures, including a Residual Network (ResNet), to classify MRI medical images across three brain cancer types
- The ResNet model, developed in Keras, achieved remarkable performance, with a balanced accuracy of 93%, a 19% increase in the balance accuracy over the baseline model. The model demonstrated strong harmonic balance between recall and precision across target labels or classes. This demonstrates the model's ability to capture most true positive instances across cancer types, while minimizing false positives across classes.

WORK EXPERIENCE

Meran Tax

Lawrence, MA

Accounting Bookkeeper

Dec 2021 – Present

- Manage the bookkeeping of 30+ small businesses from a variety of industries, including grocery stores, day care centers, restaurants, and more.
- Led a project to change our tedious manual bookkeeping process to an efficient semi-automated process. After implementation, we experienced a doubling in bookkeeping productivity. This led to meaningful labor cost savings for Meran Tax.
- Led a project to adopt OCR bank statement software for our bookkeeping process. This software adoption helped boost productivity by no longer resorting to manual entries of transactions when the file needed for semi-automation was not available.

Portsmouth Regional Hospital (HCA Health Care)

Portsmouth, NH

Staff Accountant

Aug 2023 – Apr 2025

- Responsible for posting high-dollar impact journal entries for the New Hampshire market, reconciling month-end postings, managing our physician contract-based services, and working on other periodic tasks (e.g., Annual inventory audit, petty cash audits, cost reporting, and tax reporting).
- Created and worked with sophisticated Excel spreadsheets, using Power Query, to centralize data pulled from multiple sources. This streamlined the compliance review process, facilitating faster payment processing for our providers.
- Created a Power Query Excel spreadsheet to automate the data preprocessing of our corporate intercompany journal entries across the market. This Power Query automatically organized corporate postings by type and cleaned the transactional metadata. This led to clearer transaction descriptions and increased productivity in JE preparation.

TECHNICAL SKILLS

Programming and Data: Python, SQL, NumPy, Pandas, Rioxarray, Rasterio, PostGIS, MongoDB, GeoPandas

Data Visualization: Vega-Altair, Seaborn, Streamlit, Plotly

Modeling: Sklearn, BERT, Keras, ML fairness evaluation, Time Series forecasting (ARIMA, GARCH LSTM Neural Nets), Traditional machine learning algorithmic modeling, Convolutional Networks, Causal Inference

